

BharatTrip Refund Escalations: Business Case & Solution Proposal

Executive Summary (TL;DR) BharatTrip is experiencing a spike in customer/agent refund escalations despite stable refund volumes. Our analysis identified a severe operational disconnect between Support and Finance, leading to **100 dropped tickets** and **149 short payments**. By implementing a **Single Source of Truth (SSOT)** and an **AI-Driven Data Reconciliation Workflow**, we can eliminate manual data leakage, automate agent communications, and project an 80% drop in escalations over the next quarter without adding headcount.

1. Defining the Problem

Customer and agent escalations regarding refunds are rising. The core underlying problem is a **severe operational disconnect between the Support and Finance teams**, characterized by siloed tracking systems and inconsistent definitions of refund statuses.

Key Communication Failures:

- 1. Data Leakage:** Informal requests via WhatsApp and Email bypass the tracker altogether, meaning they are never reviewed or processed.
- 2. "Closed" Status Ambiguity:** The Support team marks tickets as "Closed" once handed off to Finance, misleading the agent into believing the refund has been disbursed.
- 3. Hidden Policy Deductions:** Finance applies necessary deductions (e.g., cancellation fees) and processes payouts, but this information is never relayed back to the agent by Support, causing "short payment" escalations.

2. Data Analysis & Insights

By cross-referencing the Support_Tracker, Finance_Tracker, and Escalations logs via a deep database join, we identified the specific patterns driving the **155 total escalations**.

[image]

The Discrepancy Breakdown

Escalation Driver	Ticket Volume	Description
Dropped Handoffs	100	Logged by Support, but entirely missing from Finance tracking.
Deduction Mismatches	149	Amount promised by Support differs from amount disbursed by Finance.
Off-Tracker Leakage	24	Escalations with "no ref" originating from unlogged WhatsApp/Email.

Insight: While a naive net volume calculation (755 Support - 689 Finance) suggests only 66 missing tickets, our deep ID-to-ID database reconciliation reveals the true scale of the failure is **100 dropped tickets**, masked by duplicate records and asynchronous closures.

Future Validation

If given more time and access, I would validate the exact schema of the Support team's ticketing software to understand why the "Closed" status is automatically triggered, and I would pull the API logs to see if Finance ever sends an automated webhook back to Support.

3. Designing a Zero-Headcount Solution

To permanently bring escalations down, BharatTrip must establish a unified **Single Source of Truth (SSOT)** and automate the communication loop.

The Proposed Workflow:

1. **Unified SSOT Tracking:** Migrate both teams onto a single shared database.
 2. **AI-Powered Ingestion:** Deploy an AI agent to monitor the Support Email and WhatsApp. The AI will parse unstructured messages and log them as structured tickets, eliminating "no ref" leakage.
 3. **Automated Communication:** A system trigger automatically emails the travel agent when the status changes to `Paid`, detailing the exact payout amount and any deductions.
 4. **Strategic Trade-offs:** We automate data entry and outbound communication, but deliberately keep *Finance payout execution* and *Support's initial vetting* manual to ensure strict financial compliance.
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4. The AI Prototype: Advanced Multi-Agent Architecture

To deliver a production-ready solution, I have developed `app.py`, a zero-friction **Streamlit Web Application** showcasing an advanced Multi-Agent architecture powered by `gemini-3.5-flash`.

[image]

Secure Integration (MCP)

In a production environment, the LLM will bridge with the internal SSOT database via the **Model Context Protocol (MCP)**. This guarantees that raw database credentials are never exposed to the LLM's context window, ensuring enterprise-grade security while allowing the AI to read/write records.

The Prototype Features

- **Metrics Dashboard:** Live KPIs tracking Escalations and Discrepancies to prove ROI.
 - **HITL Workflow:** A UI for support staff to review AI-generated explanatory messages before they are sent to external agents.
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5. Measuring Success

To ensure the solution is working, we will monitor two primary metrics over the next month:

1. **Escalation Rate:** The percentage of refund requests that result in a formal escalation.
Target: >50% drop within 30 days.
 2. **Tracker Discrepancy Count:** The number of tickets marked "Closed" by Support without a matching payout record in Finance. Target: Zero.
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6. Assumptions Made

As the brief was deliberately incomplete, I proceeded with the following assumptions to build the prototype:

- **Finance is the ultimate authority on payout amounts:** If Support quotes 5000 INR but Finance pays 4500 INR, I assumed the 500 INR deduction is legitimate (cancellation fee) rather than a Finance error, meaning the fix requires better communication to the agent rather than a refund correction.
- **WhatsApp/Email formats are unpredictable:** I assumed agents will not follow a strict template, necessitating an LLM to dynamically extract the `agent_name`, `route`, and `intent` from informal natural language.